ACML101: Introduction to Chemistry
Inorganic Chemistry: Tutorial 3

Few questions on coordination chemistry and then from organometallic chemistry

Few reading slides on organometallic chemistry has been given before starting the tutorial

18 electron rule : How to count electrons

The rule states that thermodynamically stable transition metal organometallic compounds are formed when the sum of the metal valence electrons and the electrons conventionally considered as being supplied by the surrounding ligands equals 18.

In general, the conditions favouring adherence to the 18 electron rule are, an electron rich metal (one that is in a low oxidation state) and ligands that are good π-acceptors

The *hapto symbol*, ***η***, with a numerical superscript, provides a topological description by indicating the connectivity between the ligand and the central atom. For example, if all the five carbon atoms of a cyclopentadienyl moiety are equidistant from a metal atom, we term it as η^5-cyclopentadienyl

Examples:

η^1-R, η^1-Ar η^2-C_2R_4 η^1-allyl, η^3-allyl, η^4- Cb, η^5-Cp, η^6-C_6H_6 η^8-C_8H_8

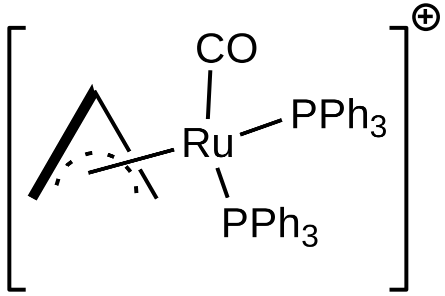
The symbol ***μ*** indicates bridging normally we have μ_2 and rarely μ_3 bridging

Examples: μ_2-CO, μ_3-CO, μ_2-CH_3, μ_2-H, μ_2-Cl, , μ_3-Cl, μ_2-OR, μ_2-PR_2, μ_2-NR_2

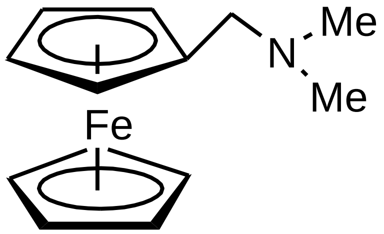
Methods of counting: Neutral atom method & Oxidation state method

Ligand	Neutral atom	Oxidation state		Ligand	Neutral atom	Oxidation state	
		Electron contribution	Formal charge			Electron contribution	Formal charge
Carbonyl (M–CO)	2	2	0	Halogen (M–X)	1	2	–1
Phosphine (M–PR_3)	2	2	0	Alkyl (M–R)	1	2	–1
Amine (M–NR_3)	2	2	0	Aryl (M–Ar)	1	2	–1
Amide (M–NR_2)	1	2	–1	acyl (M–C(O)–R	1	2	–1
Hydrogen (M–H)	1	2	–1	η^1-cyclopentadienyl	1	2	–1
Alkene (sidewise) η^2-	2	2	0	η^1-allyl	1	2	–1
Alkyne (sidewise) η^2-	2	2	0	η^3-allyl	3	4	–1
η^2-C_{60}	2	2	0	η^5-cyclopentadienyl	5	6	–1
Nitrosyl bent	1	2	–1	η^6-benzene	6	6	0
Nitrosyl linear	3	2	+1	η^7-cycloheptatrienyl	7	6	+1
Carbene (M=CR_2)	2	4	–2	Carbyne (M≡CR)	3	6	–3
Alkoxide (M–OR)	1	2	–1	Thiolate (M–SR)	1	2	–1
μ-CO (M–(CO)–M)	2	2	0	μ-H	1	2	–1
μ-alkyne	4	4	0	μ-X (M–X–M) X = halogen	3	4	–1
μ-alkyl	1	2	–1	μ-amido (M–(NR_2)–M	3	4	–1
μ-phosphido (M–(PR_2)–M	3	4	–1	μ-alkoxide (M–(OR)–M	3	4	–1

Examples:



	neutral atom method	oxidation state method
Ru	8	6 (Ru +2)
η^3- allyl	3	4
2 PPh_3	4	4
CO	2	2
charge	-1	not required
	16	16



	neutral atom method	oxidation state method
Fe	8	6 (Fe +2)
2 η^5-Cp	10	12
	18	18

Neutral atom method: Metal is taken as in zero oxidation state for counting purpose

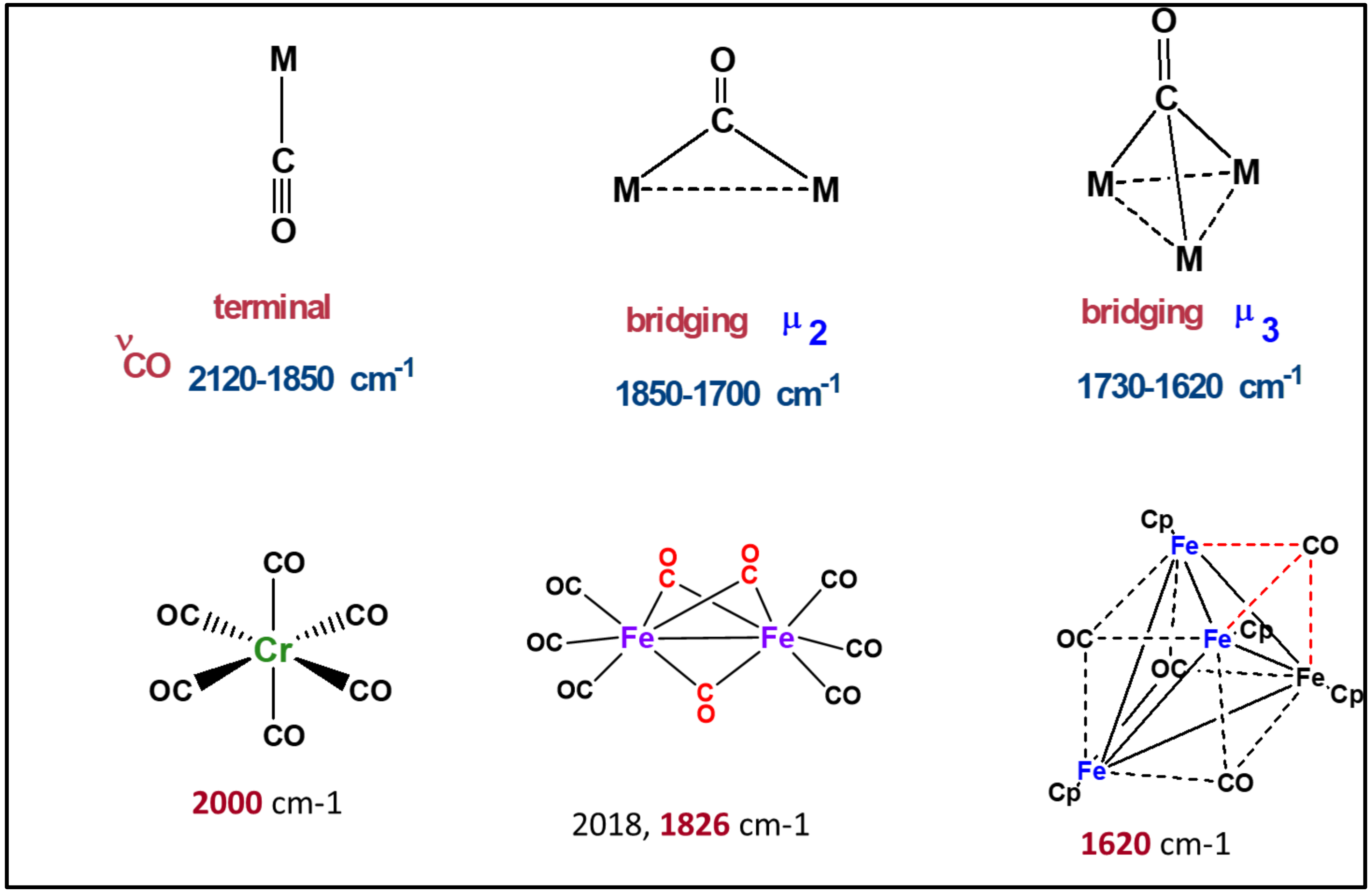
Oxidation state method: We first arrive at the oxidation state of the metal by considering the number of anionic ligands present and overall charge of the complex

Suggestion: Focus on one counting method till you are confident

Easy way to remember ligand electron contribution for neutral atom counting method

		Electron contribution
Neutral terminal :	**CO, PR_3, NR_3**	**2 electrons**
Anionic terminal :	**X-, H-, R-, Ar-, R_2N-, R_2P-, RO-**	**1 electron**
Hapto ligands	**: η^2-C_2R_4 η^2-C_2R_2, η^4-C_2R_2 ,η^1-allyl, η^3-allyl, η^4- Cb, η^5-Cp, η^6-C_6H_6 η^7-C_7H_7 η^8-C_8H_8 η^2-C_{60}, η^5-R_5C_{60}**	**same as hapticity**
bridging neutral	**μ_2-CO, μ_3-CO**	**2 electrons**
Bridging anionic	**μ_2-CH_3, μ_2-H (no lone pairs)**	**1 electron**
Bridging anionic (with 1 lone pair)	**μ_2-Cl, , μ_2^-OR, μ_2-PR_2, μ_2-NR_2**	**3 electrons**
μ_3-Cl(2 l.p)		**5 electrons**
Bridging alkyne		**4 electrons**
NO linear		**3 electrons**
NO bent (l. p on nitrogen)		**1 electron**
Carbene M=C		**2 electron**
Carbyne M≡C		**3 electron**

Terminal versus bridging carbonyls



Counting the electrons helps to predict stability of metal carbonyls. But it will not tell you whether a CO is bridging or terminal

Others factors which affect ν_{CO} stretching frequencies:

1. Charge on the metal
2. Effect of other ligands

$$M-C\equiv O \longleftrightarrow M=C=O$$

ν CO Higher

ν CO Lower

More back bonding

Oxidative addition

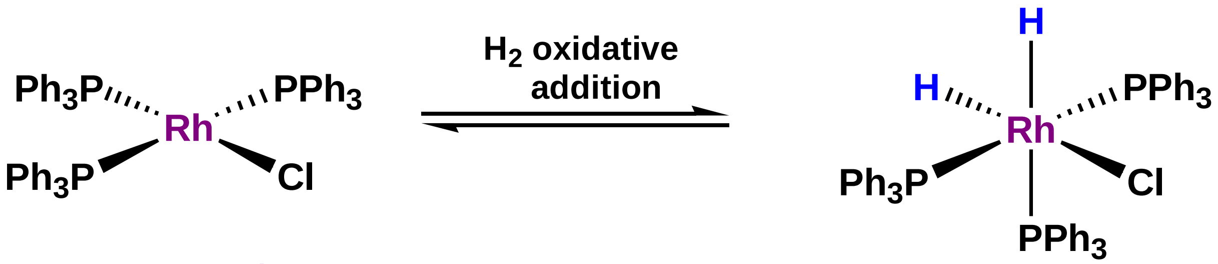
When addition of ligands is accompanied by oxidation of the metal, it is called an oxidative addition reaction

$$\underset{d^n}{L_nM} + XY \longrightarrow \underset{d^{n-2}}{L_n(X)(Y)M}$$

OX state of metal increases by 2 units

Coordination number increases by 2 units

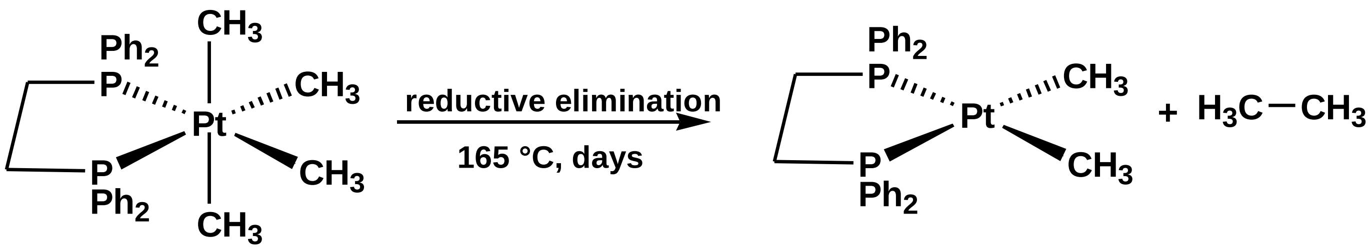
2 new anionic ligands are added to the metal



Rh^{+1} Rh^{+3}

Reductive elimination

Almost the exact reverse of Oxidative Addition



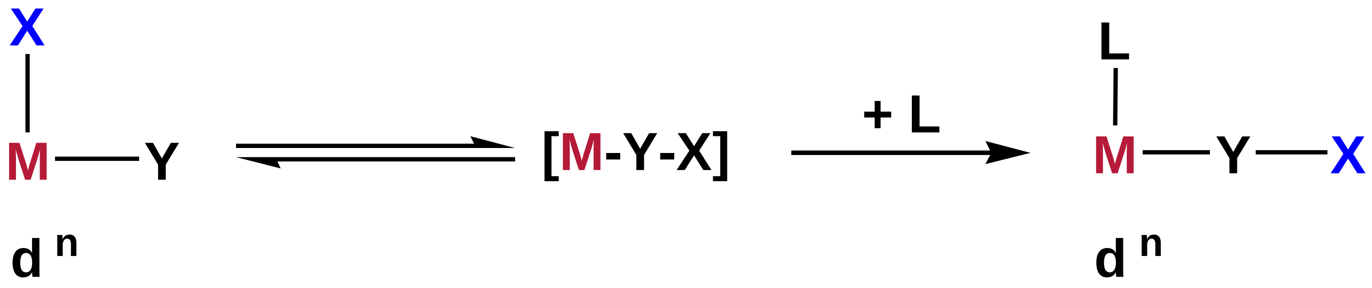
Pt^{4+} Pt^{2+}

Oxidation state of metal decreases by 2 units

Coordination number decreases by 2 units

2 cis oriented anionic ligands form a stable σ bond and leave the metal

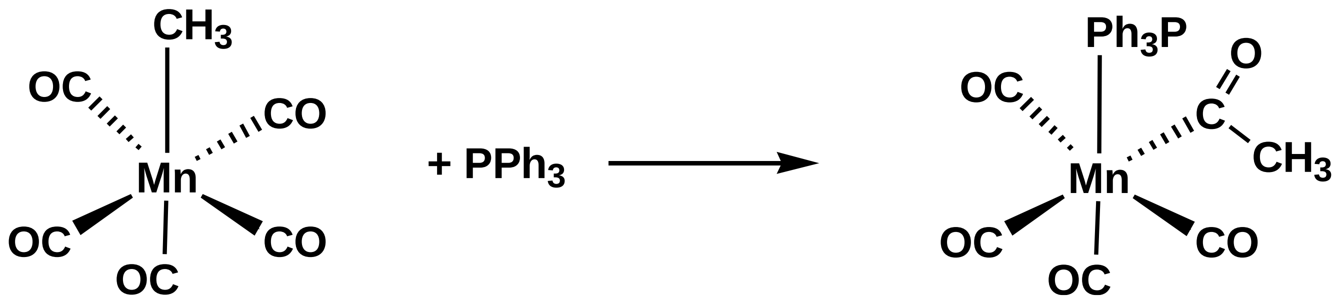
Migratory Insertion



No change in the formal oxidation state of the metal

A vacant coordination site is generated during a migratory insertion (*which gets occupied by the incoming ligand*)

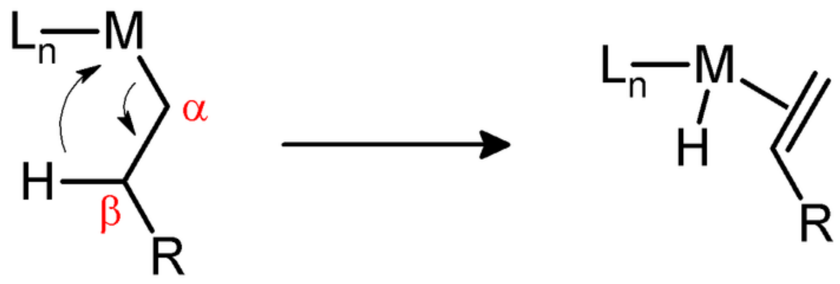
The groups undergoing migratory insertion must be *cis* to one another



These reactions are enthalpy driven and although the reaction is entropy prohibited the large enthalpy term dominates

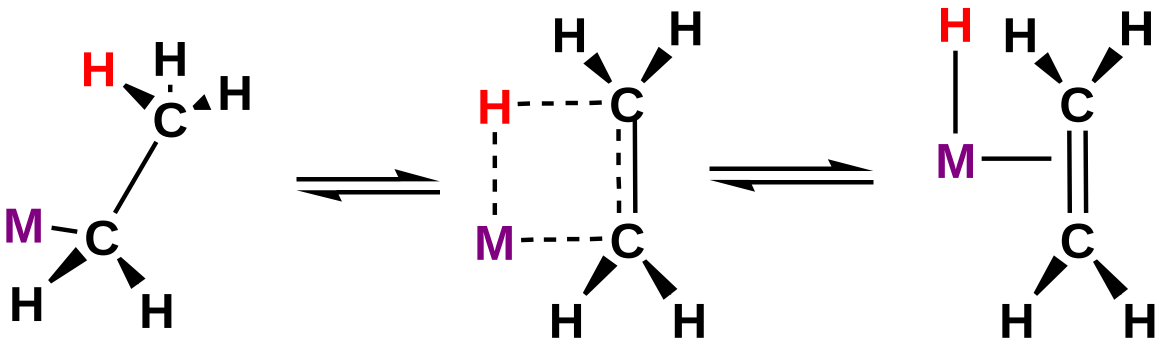
β-Hydride elimination

Beta-hydride elimination is a reaction in which an alkyl group having a β hydrogen, σ bonded to a metal centre is converted into the corresponding metal-bonded hydride and a π bonded alkene. The alkyl must have hydrogens on the beta carbon. For instance butyl groups can undergo this reaction but methyl groups cannot. The metal complex must have an empty (or vacant) site cis to the alkyl group for this reaction to occur.



No change in the formal oxidation state of the metal

mechanism



β-hydrogen elimination does not happen when

- the alkyl has no β-hydrogen (as in $PhCH_2$, Me_3CCH_2, Me_3SiCH_2)
- (ii) the β-hydrogen on the alkyl is unable to approach the metal (as in C≡CH)
- the M–C–C–H unit cannot become coplanar

β-hydrogen elimination can either be a vital step in a reaction or an unwanted side reaction

(1) Which of the following complexes follow 18 electron rules

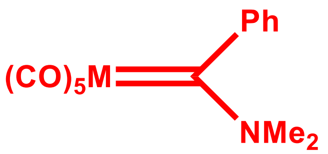
(a) $[Cp_2Co]$ (b) $[Cp_2Co]^+$ (c) $V(CO)_6$ (d) $[V(CO)_6]^-$ (e) $MeMn(CO)_5$ (f) $(\eta^3\text{-}C_3H_5)_2Rh(\mu\text{-}Cl)_2Rh(\eta^3\text{-}C_3H_5)_2$

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg

(2) Find the unknown quantity, if the central metal ion follows 18 electron rule.

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg

(a) $Re(CO)_nBr$ find n?

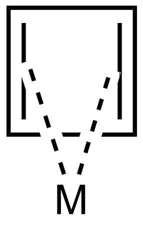
(b) Find M (1^{st} row) in 

?

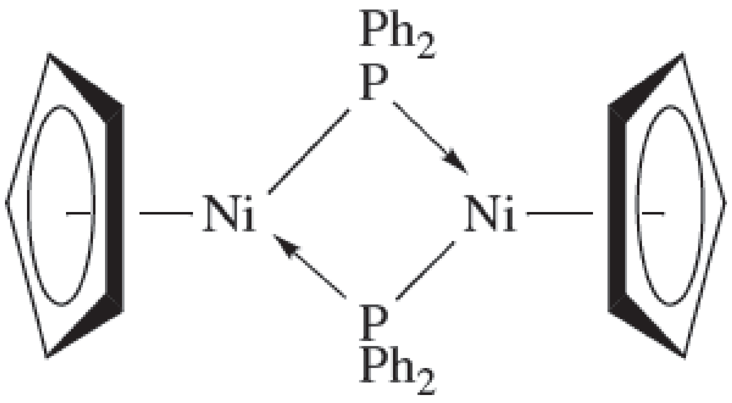
(c) $[Os(CO)_4]^z$ find z?

(d) $(\eta^x\text{-}C_7H_8)Mo(CO)_3$ find x?

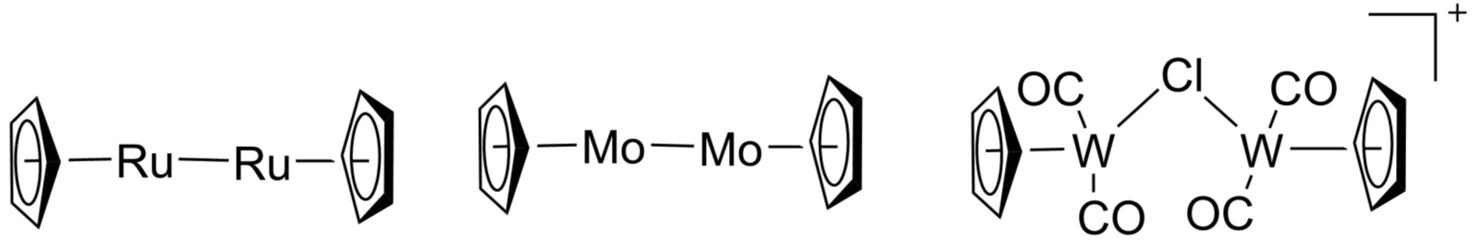
(e) $[(Cp)Co(C_nH_n)]$



(3) Solely on the basis of the 18-electron rule, suggest whether (η^5-Cp)Ni(μ-PPh_2)$_2$Ni(η^5-Cp) might be expected to contain a metal–metal bond (Ni^0 = [Ar]$3d^84s^2$).



(4) Four chlorine ligands are missing in each of the given skeletons of dimeric compounds a, b and c. Given that all of them obey the 18-electron rule and no additional metal-metal bonds are present, attach the missing Cl ligands on the complexes in the appropriate manner.



(a) (b) (c)

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg

Answer:

(5) Reaction of $Mn_2(CO)_{10}$ with Br_2 resulted in an orange colored powdery compound A. On keeping A under high vacuum in a sealed tube at room temperature for several days resulted in the formation of dark orange crystals of a new compound B. Analysis of A and B indicated only infra red bands in the range of 2045-1950 cm^{-1}. A and B are stable compounds and both contain bromine atoms. Draw the structures of A and B.

(6) Which case of $MoL_3(CO)_3$ is expected to exhibit a lower CO stretching frequency; L = PF_3, or L = PMe_3? Explain

(7) Arrange the following in accordance with decreasing carbonyl stretching frequency ($\nu_{C\text{-}O}$) in cm^{-1}, $Ni(CO)_4$, $[Co(CO)_4]^-$ and $[Fe(CO)_4]^{2-}$

(8) Classify the following reactions as oxidative addition, reductive elimination, migratory insertion, β-hydrogen elimination, ligand coordination change or simple addition. (Check the change of oxidation state and coordination number)

21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg

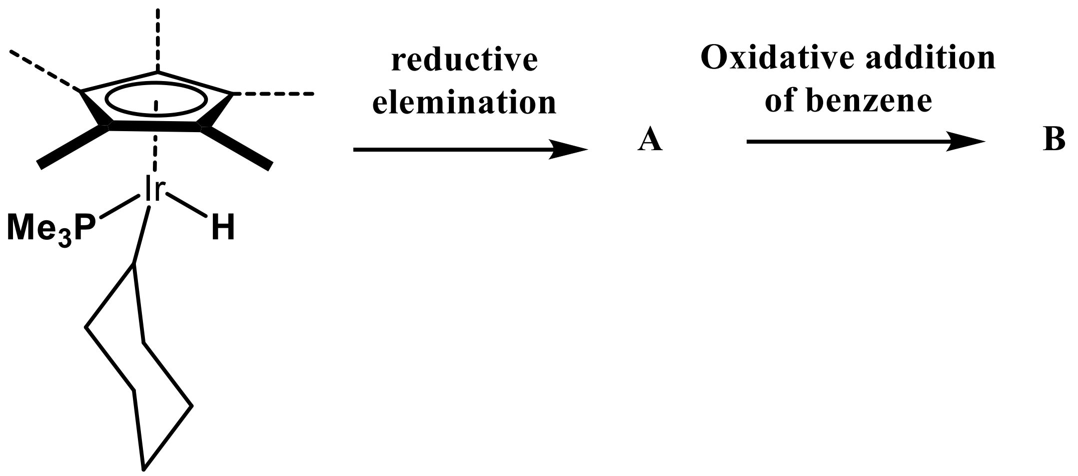
(a) $[RhI_3(CO)_2CH_3]^- \rightarrow \{RhI_3(CO)(solvent)[C(O)CH_3]\}^-$

(b) $Co_2(CO)_8 + H_2 \rightarrow 2\ HCo(CO)_4$

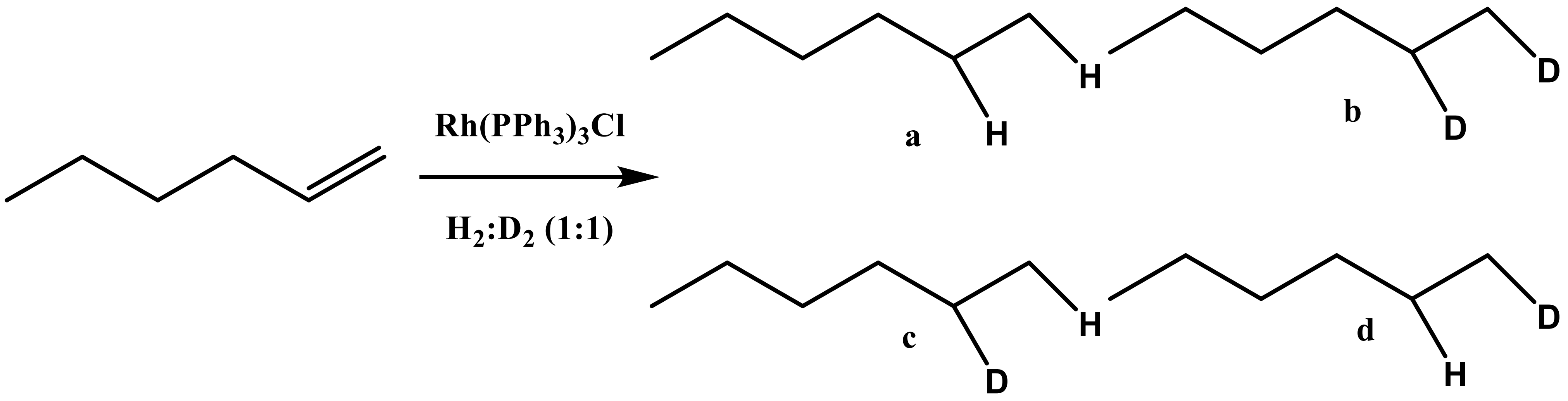
(c) $Ir(PPh_2Me)_2(CO)Cl + CF_3I \rightarrow Ir(I)(CF_3)(PPh_2Me)_2(CO)Cl$

(d) $TiCl_4 + 2\ Et_3N \rightarrow TiCl_4(NEt_3)_2$

(9) Predict the product of the following reaction? Count the change of oxidation state and valence shell electrons of the metal center. (Ir = [Xe]$5d^7 6s^2$)(previous year question)

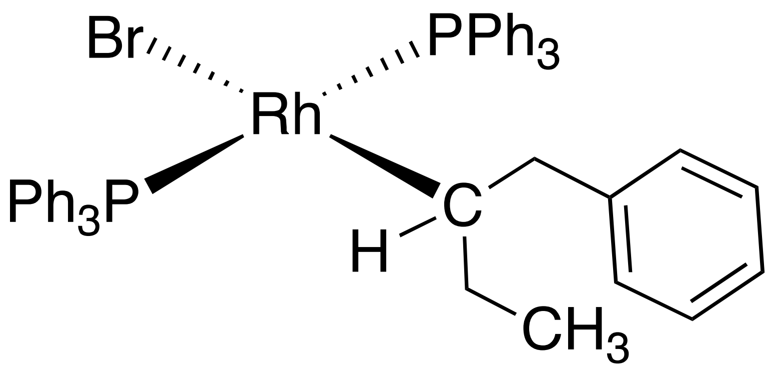


10. Which are the likely product/products in the following reaction?



11. Explain why trans-[Pt(Et)(Cl)(PEt_3)$_2$] readily decomposes, whereas trans-[Pt(Me)(Cl)(PEt_3)$_2$] does not.

12. The following Palladium complex is not stable, undergoes rearrangement reaction, results two different products. Draw the structure of two products.



Answer:

13. The complex $(PPh_3)_3Rh(Me)$ is cleaved in the presence of D_2 and forms methane and $(PPh_3)_2Rh(D)(PPh_2C_6H_4D)$. Suggest a suitable mechanism for the reaction. (Hints: 1^{st} step is intramolecular oxidative addition)

Answer